05 Calling functions: sort, sum, length, mean, median

Use R's built-in functions to sort and summarize a list of numbers.

BY THE END OF THIS SECTION YOU WILL BE ABLE TO

- Call a function with values in brackets
- Summarize numbers with sum, length, mean and median
- Understand why mode() gives a surprise

The R code in this lesson

```
x <- c(1,5,3,5,6,4,3,65,3,3,4,6,8,5433,45)
sort(x)     # smallest to largest
sum(x)      # 5594
length(x)   # 15 values
mean(x)     # 372.93
median(x)   # 5
mode(x)     # "numeric": mode() tells the type, not the most common value
```

What you will hear (times count from the end of the intro)

Intro Functions are the heart of R. In this lesson, you will use five friendly functions to sort and summarize numbers, and you will see one function that does something surprising. Let's explore them together, slowly.

0:00 Here are the functions from our list. First, sort. It puts our numbers in order, from the smallest to the largest. In the console, we see one, three, three, all the way up to five thousand four hundred thirty three.

0:15 Next, sum. Notice, sum does not even need c. You can give it the numbers directly. The total is five thousand five hundred ninety four.

0:30 Now length. Length counts how many values we have. I type length, open the brackets, and inside, c with the same numbers.

1:00 Run it. The answer is fifteen. We have fifteen numbers.

1:20 Next, mean, which is the average. And median, the middle value when the numbers are sorted.

1:35 When I first ran mode, I got an error: unused arguments. That is because I forgot c around the numbers. So I add c, and run it again.

2:05 The mean is three hundred seventy two point nine three. That is high, because one number, five thousand four hundred thirty three, pulls the average up.

2:20 The median is only five. When a few values are very big, the median gives a better picture of the typical value. That is a useful lesson for any data work.

2:30 And mode? It says numeric. In R, mode tells you the type of the data, not the most common value. A small surprise, and a good one to remember.

2:48 Finally, I scroll back to the top of our script. Look how much we wrote today. Calculations, data types, variables, vectors, comparisons, and functions. Excellent work!